

Reducing urgent waits without overloading the clinic

A discrete-event simulation study of three appointment policies

48.8%

lower urgent mean
under Q3

Routine mean rises
to 9.053 days

At 98% nominal load, priority must be created rather than found

0.98

nominal load

Demand almost matches capacity, so every protection decision redistributes delay.

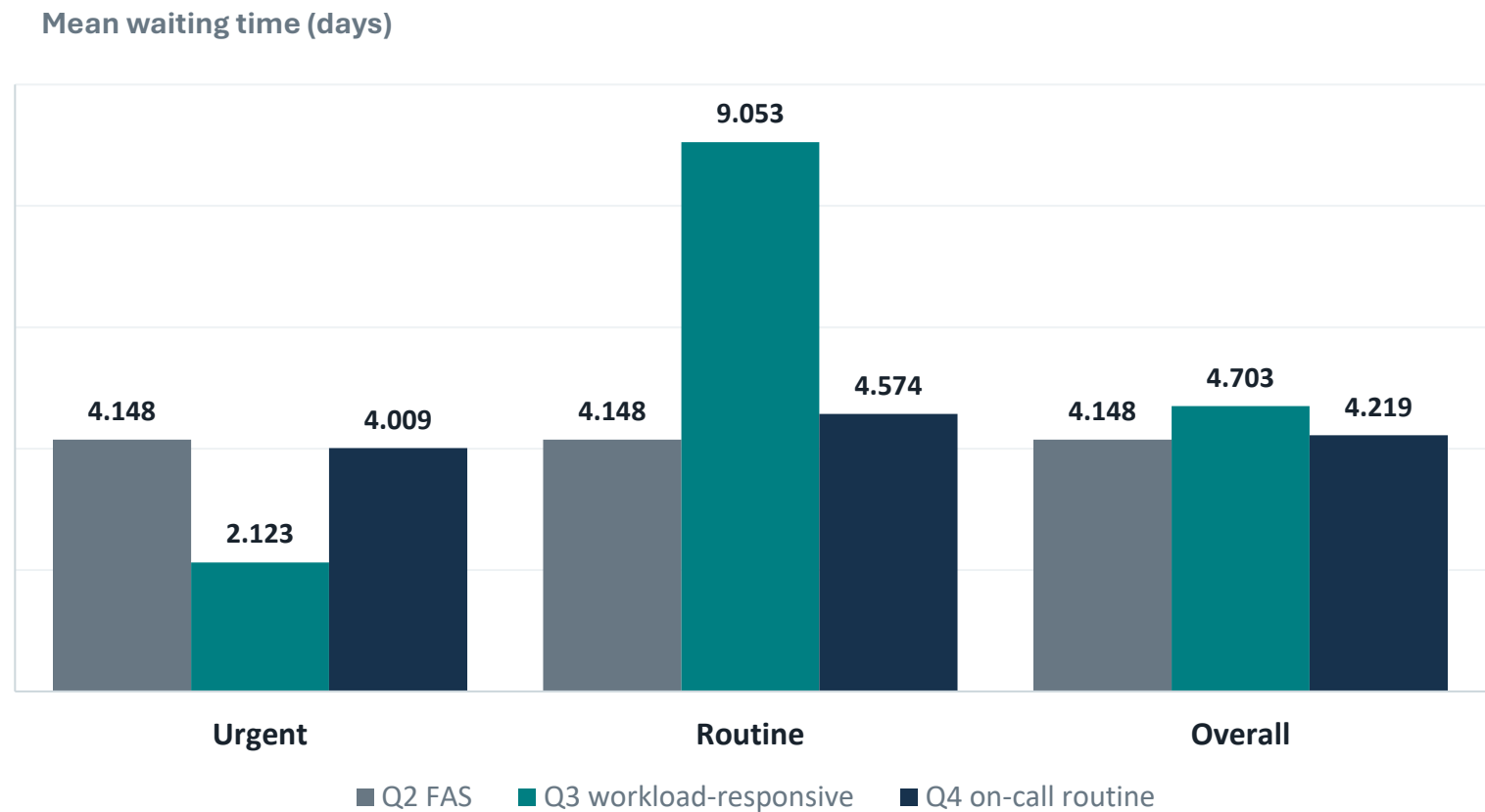
Urgent demand	3.69/day
Routine demand	2.19/day
Available capacity	6 slots/day

Literature framing

Capacity protection can improve urgent access, but the cost is displaced delay for flexible or lower-priority patients.

Which rule best converts scarce flexibility into shorter urgent waits?

Q3 wins the stated objective, but not every service outcome



Decision

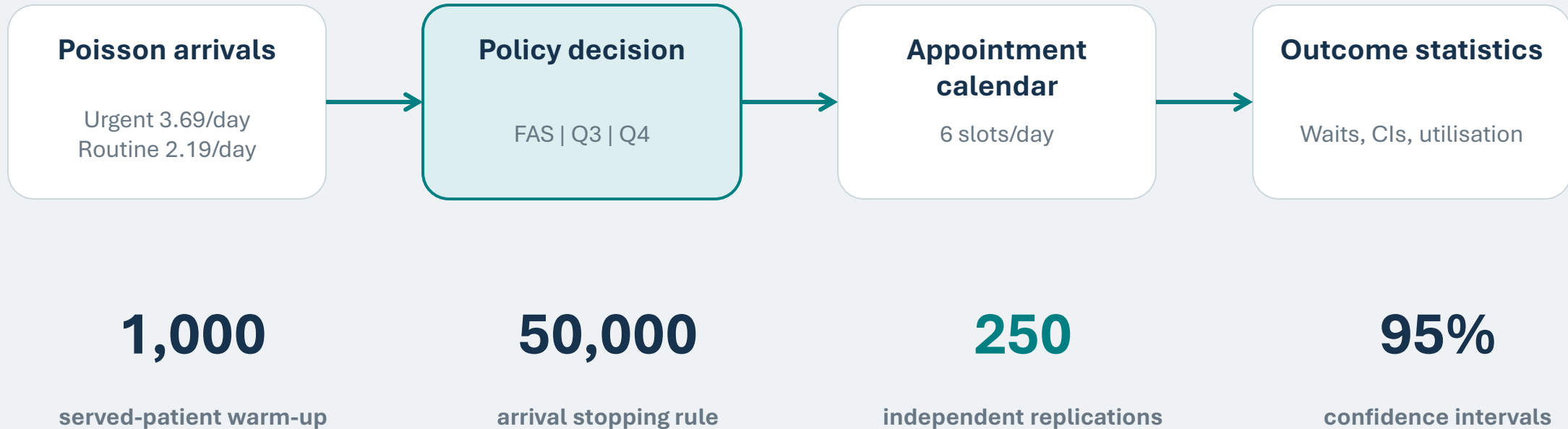
Choose Q3

Largest urgent reduction while overall mean remains feasible

Service risk

Routine mean rises to 9.053 days

A common experiment design makes the three policies comparable



Precision target met: relative CI half-width $\leq 5\%$ for every selected policy

Q2 provides a stable benchmark rather than a priority solution

4.148

overall mean waiting (days)

95% CI [3.978, 4.318]

Urgent 4.148 | Routine 4.148

Warm-up sensitivity

500 served	4.133
1,000 served	4.148
2,000 served	4.172

Strongly overlapping confidence intervals support a stable benchmark.

Q3 halves urgent waiting by shifting delay to routine patients

Postponement = round(α_1 NU + α_2 NU²)

$\alpha_1 = 0$ | $\alpha_2 = 0.35$

Urgent backlog increases protection nonlinearly.

2.123

urgent mean (days)

48.8% below FAS

Routine

9.053 days

Overall

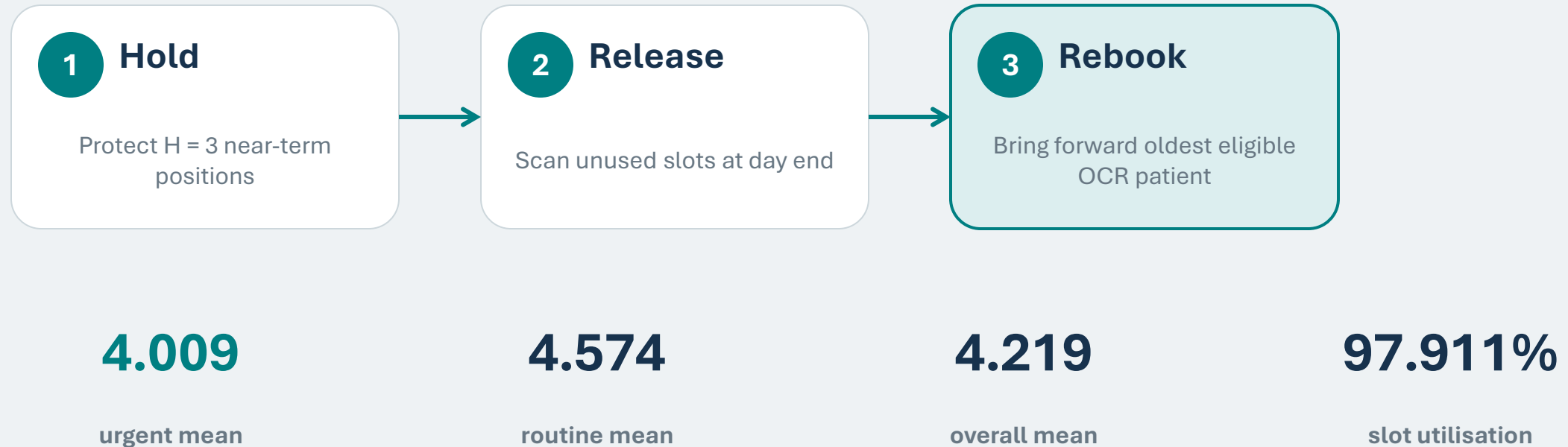
4.703 days

Feasibility

Upper CI 4.893

Q4 preserves utilisation but creates little urgent capacity

$\eta = 0.2$ | $L = 0$ | $H = 3$



0.515 days saved per rebooked OCR patient

Recommend Q3, with a routine service-level guardrail

Choose Q3

Urgent mean 2.123 days
Overall upper 95% CI 4.893 days

Guardrail required

Routine mean = 9.053 days
Add a routine service limit.

What the evidence supports

Q3 is the best tested policy for the stated objective. It is not a free efficiency gain; it redistributes waiting from urgent to routine patients.

Next experiment

Combine Q3's backlog signal with controlled OCR release, then test a routine service-level constraint.

Limitations: stationary demand, deterministic service, no no-shows or overtime, finite parameter search.

Full 95% confidence intervals for selected policies

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Metric	Mean	95% confidence interval
Q2 overall	4.148	3.978-4.318
Q3 urgent	2.123	2.075-2.172
Q3 routine	9.053	8.621-9.484
Q3 overall	4.703	4.513-4.893
Q4 urgent	4.009	3.862-4.156
Q4 routine	4.574	4.426-4.722
Q4 overall	4.219	4.072-4.367
Q4 utilisation	97.911%	97.858%-97.965%

FAS is insensitive to the tested warm-up lengths

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Warm-up	Overall mean	95% confidence interval
500 served	4.133	3.965-4.302
1,000 served	4.148	3.978-4.318
2,000 served	4.172	3.999-4.345

The intervals overlap strongly; 1,000 served balances transient removal and data retention.

Q3 is selected at the feasible boundary

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Selected setting

$\alpha_1 = 0$ | $\alpha_2 = 0.35$

Best tested feasible boundary

2.123

urgent mean

9.053

routine mean

4.703

overall mean

The finite screen identifies the best tested setting, not a global optimum.

Q4 screening rules out aggressive protection

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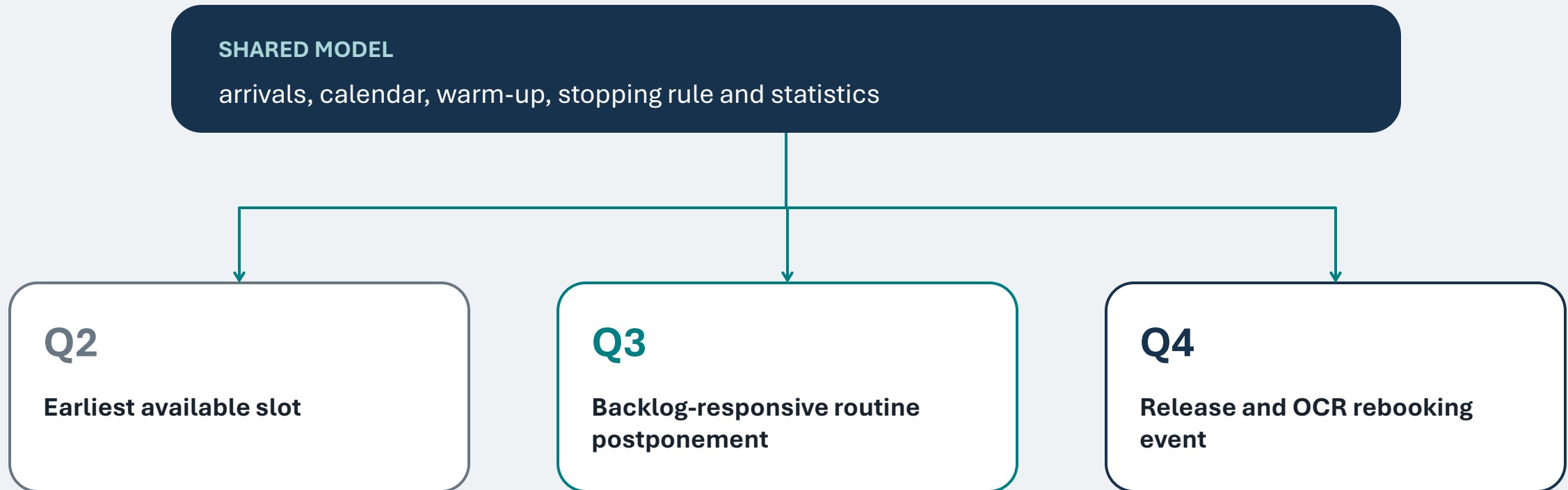
Policy	Urgent	Routine	Overall
eta 0.1, H 3	4.107	4.691	4.325
eta 0.1, H 2	4.109	4.323	4.189
eta 0.2, H 3	4.009	4.574	4.219
eta 0.2, H 2	4.093	4.304	4.172

H = 3, eta = 0.2 is the direct AnyLogic validation; other rows are event-equivalent screening estimates.

H = 4 was too aggressive under 98% load.

The AnyLogic models differ only at the scheduling decision

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Independent replications use explicit random seeds and the same output definitions.

How the literature shaped the model

- Reviews: Cayirli & Veral (2003); Gupta & Denton (2008); Ahmadi-Javid et al. (2017)
- Capacity protection: Dobson et al. (2011); Creemers et al. (2012)
- Dynamic scheduling: Patrick et al. (2008); Keyvanshokoo et al. (2021)

AI assistance

AI assistance supported drafting, formatting and checking. Model choices, simulation runs, interpretation and final responsibility remain with the student.

Full bibliography: nine peer-reviewed sources in the submitted report.